

Numerical methods : numerical lab evaluation

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Allotted time : 1H15 minutes ; (1/3 time 25 minutes more)

Project Instructions

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| Not reading all the instructions below = penalty of 20/20 pts. |
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- **Comments** Add comments to explain the code !
- **Parameter Values** Never use hard-coded parameter values (like $a = 2$ and then use "2"; use systematically "a" instead of "2" once "a" is assigned).
- **Formula References** If possible, refer to formulas from the handout (formula number or page).
- **Understandability** The entire project should be understandable by a colleague (M1 level).
- **Difference with class corrections** for implementation choices different than what has been seen in class for the same question explain why you took that choices.
- **Code Cleanliness** Python code should be as clear and efficient as possible: **avoid "for" loops unless absolutely necessary**, etc. Write the shortest code possible (that is still clear and easy to understand).
- **Execution Requirements** The program should run without any configuration or intervention (watch out for special libraries, etc). Do not use special packages except the standard ones used in classroom.
- **Project Start and name** The program should start with a comment containing **the name(s) of the author(s) and the chosen project**

and the date. Absence of this comment = 10p penalty. The name of the file should be the name of the author and the subject code of the project (see title of each project) separated by underscore '_'.

- **Submission of the code :** by Teams in a private message; do not modify the project afterwards. Any **modification of the code in teams after the deadline = PENALTY 20/20 points ! PENALTY 15/20** if sent in a format other than '*.py' or '*.ipynb' (teams)!!!
- **Project choice :** choose only one project and **send a teams message with your choice at the latest 20 minutes after the start** of the evaluation session. Do not modify this message at any time (penalty 10/20 pts). Your choice is final.

1 ODE System, code ode_regime

We consider a compartmental ODE as in the epidemic models of the course. Here we have two sets of classes, the X_j classes, $j \leq J$ and Y_j classes; the ODEs are :

$$\frac{dX_j}{dt} = -\lambda_j X_j + p(1-m) q_j L_X, \quad X_j(0) = X_{j0} \quad (1)$$

$$\frac{dY_j}{dt} = pm \lambda_j X_j + g Y_j, \quad Y_j(0) = 0 \quad (2)$$

$$L_X(t) = \sum_k \lambda_k X_k(t) \quad (3)$$

Contraint: simulation will need to ensure all classes remain positive at all times.

Parameters : $T = 300$, $J = 10$, $(\lambda_j)_j = 10^{\text{np.linspace}(-6,3,J)}$, $q_j = 1/J$, $p = 0.95$, $m = 10^{-8}$, $g = 0.1$, initial value $X_{init} = 10^6$, $X_{j0} = X_{init}/J$.

To plot : evolution of $\sum_a X_a(t)$, $\sum_a Y_a(t)$, $\frac{X_j(t)}{\sum_a X_a(t)}$, same for Y classes; histogram of classes X_j with respect to j , same for Y_j .

Comparison: compare the dynamics with several situations:

a/ when we have $g\sqrt{Y_j}$ instead of gY_j in the ODE of Y_j .

a/ when we have $dY_j = (pm\lambda_j X_j + g\sqrt{Y_j})dt + \sqrt{g\sqrt{Y_j}}dW_t^j$ instead the ODE of Y_j . Here W_t^j are independent Brownian motions. Be careful, all X_j and Y_j have to remain positive.

Indication : if necessary can use Euler-Maruyama scheme for $Z_j = Y_j^{1/4}$.

Comparison: by changing all parameters of the model find whether exists a regime of parameters where histogram of Y_j is unimodal i.e. maximum density in the middle (not necessarily symmetric).

2 VP Diffusion SDE vs Backward Probability-Flow ODE (1D), code `vp_sde_ode`

We consider a *variance-preserving* (VP) diffusion in 1D and its corresponding *probability-flow ODE*. The purpose is to simulate:

- (i) the **forward SDE** with Euler–Maruyama,
 - (ii) the **backward ODE** with RK4/ Heun (vectorized) or `odeint` (non-vectorized, penalty),
- and compare, at n_T intermediary times, the empirical marginal distributions by overlaid histograms.

Target (initial) distribution

Let the initial law p_0 be the (normalized) sum of two indicator functions:

$$p_0(x) = \frac{1}{2} \mathbf{1}_{[-2,-1]}(x) + \frac{1}{2} \mathbf{1}_{[1,2]}(x). \quad (4)$$

Forward VP diffusion (SDE) Fix a constant schedule (step) $\beta > 0$ and consider the VP SDE:

$$dX_t = -\frac{\beta}{2} X_t dt + \sqrt{\beta} dW_t, \quad X_0 \sim p_0. \quad (5)$$

This SDE has the explicit conditional law

$$\text{law}(X_t | X_0 = x_0) \sim \mathcal{N}(a(t)x_0, s^2(t)), \quad a(t) := e^{-\beta t/2}, \quad s^2(t) := 1 - e^{-\beta t}. \quad (6)$$

Analytic expressions¹ for $p_t(x)$ and $\partial_x \log p_t(x)$: Using (6) and (4), for $t > 0$ the marginal density p_t is:

$$p_t(x) = \frac{1}{2a(t)} \left[\Phi\left(\frac{x+2a(t)}{s(t)}\right) - \Phi\left(\frac{x+a(t)}{s(t)}\right) + \Phi\left(\frac{x-a(t)}{s(t)}\right) - \Phi\left(\frac{x-2a(t)}{s(t)}\right) \right], \quad (7)$$

$$p'_t(x) = \frac{1}{2a(t)s(t)} \left[\varphi\left(\frac{x+2a(t)}{s(t)}\right) - \varphi\left(\frac{x+a(t)}{s(t)}\right) + \varphi\left(\frac{x-a(t)}{s(t)}\right) - \varphi\left(\frac{x-2a(t)}{s(t)}\right) \right]. \quad (8)$$

¹**Note:** you are NOT requested to prove any of the formulas (7), (8), (9), doing so will result in a penalty.

where Φ is the standard normal CDF and $s(t) = \sqrt{1 - e^{-\beta t}}$ and φ the standard normal PDF. Note that (8) is obtained from (7) by differentiation. Hence we know explicitly the quantity $\partial_x \log p_t(x)$ which is called the ‘score’:

$$\partial_x \log p_t(x) = \frac{p'_t(x)}{p_t(x)}, \quad t > 0. \quad (9)$$

Backward probability-flow ODE To the forward SDE (5) we associate some ODE:

$$Y'_t(\omega) = -\frac{\beta}{2} Y_t(\omega) - \frac{\beta}{2} \partial_y \log p_t(Y_t(\omega)), \quad t \in (0, T]. \quad (10)$$

To run it *backward*, integrate (10) from $t = T$ down to $t = \varepsilon$ (small $\varepsilon > 0$) using the time grid (t_n) in decreasing order. Y_t is written as a random variable because the value Y_T is not fixed but is sampled from a standard normal variable. Each sampled value results in another ODE trajectory; we are interested in the distribution of the $Y_t(\omega)$ for t in some temporal grid.

Note (avoid $t = 0$). Since $s(t) \rightarrow 0$ as $t \downarrow 0$, evaluate the ODE only for $t \geq \varepsilon$ (e.g. $\varepsilon = 10^{-3}$) when computing (9).

Parameters Example choices: $T = 2$, $\beta = 1$, $n_{SDE} = 10^5$ (the number of SDE trajectories : adjust if necessary), $n_T = 10$ (number of snapshot times) $\Delta t = 10^{-3}$ (SDE time step); snapshot times: $0 < t_1 < \dots < t_{n_T} = T$ (e.g. log-uniform); $n_{ODE} \geq 5 \times 10^3$ (the number of ODE trajectories : adjust if necessary),

What to do

1. Forward SDE simulation (Euler–Maruyama).

Simulate (5) with step Δt for n_{SDE} trajectories. Store samples $\{X_{t_m}^{(k)}\}_{k=1}^{n_{SDE}}$ at each snapshot time t_m , $m = 1, \dots, n_T$.

2. Backward ODE simulation.

Sample Y_T at terminal time; for each randomness ω_ℓ , solve (10) backward from T to ε Store samples at the same snapshot times $\{t_m\}$.

Ideally you should use a RK4 or Heun scheme, **VECTORIZED** to advance all ODEs at the same time. Otherwise use `odeint` non-vectorized (penalty).

3. Comparison by marginal snapshots.

For each $t_m \in \{0, t_1, \dots, t_{n_T} = T\}$, plot on the same figure:

- a histogram (density=True) of the forward SDE samples $\{X_{t_m}^{(k)}; k \leq n_{SDE}\}$,
- a histogram (density=True) of the ODE samples $\{Y_{t_m}(\omega_\ell); \ell \leq n_{ODE}\}$,

with transparency (alpha) so both are visible. Use identical bins and identical x -range for the two histograms.

Required / questions:

- Code implementing (5) (Euler–Maruyama), (7)–(9), and (10).
- A grid of n_T plots overlaying the two histograms at each snapshot time.
- What do you observe ? Can you explain theoretically ?

3 No programming : correction of exercises in LaTeX (subject code: ‘Corr_exo’)

Attention: because there is no numerical programming here maximum mark is 15/20.

Write a correction **in LaTeX** for the following exercises : ‘Existence of Implicit Schemes’, ‘Euler-Lagrange multipliers: finding the derivative’ and ‘Multi-step SDE Scheme’. Each is worth 5 points.

Penalty for not writing in LaTeX = 15points.

Exercise 3.1 (Existence of Implicit Schemes). Let $\Psi : \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be Lipschitz with respect to all arguments, and let $h > 0$.

1. Show that the equation

$$y = x + h\Psi(t, x, y), \quad (11)$$

has a unique solution for small enough h and provide a numerical method to compute it. Hint: Picard iterations can be used. Notation: the solution will be denoted by $y = s(t, x, h)$.

2. Now, let y be a solution of (11), and ϕ be the function defined by

$$y = x + h\phi(t, x). \quad (12)$$

Provide the formula for ϕ in terms of $s(\cdot)$ and $\Psi(\cdot)$ and show that ϕ is well-defined and Lipschitz for sufficiently small h .

Exercise 3.2 (Euler-Lagrange multipliers: finding the derivative). Let $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be C^2 functions. Suppose that the equation $g(x, y) = 0$ has a unique solution $y = \mathcal{Y}(x)$ when x stays in some neighborhood of x_0 , with $\mathcal{Y}$ being C^1 . Denote $y_0 = \mathcal{Y}(x_0)$ and assume $\nabla_y g(x_0, y_0) \neq 0$. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ defined in a neighborhood of x_0 as $F(x) = f(x, \mathcal{Y}(x))$. We study a situation where $\mathcal{Y}(\cdot)$ is difficult to obtain but we want nevertheless to compute the gradient $\nabla_x F(x)$ without using $\mathcal{Y}(\cdot)$. We introduce the Lagrangian $L(\lambda, x, y) = f(x, y) + \lambda g(x, y)$.

1. Show that there exists λ_0 such that :

$$\nabla_y L(\lambda_0, x_0, y_0) = \nabla_\lambda L(\lambda_0, x_0, y_0) = 0. \quad (13)$$

2. Show that : $\nabla_x F(x_0) = \nabla_x L(\lambda_0, x_0, y_0)$.

3. Explain how this can be used to compute the gradient for: $f(x, y) = x + y$, $g(x, y) = x^2 + y^2 - 1$ in some point (x_0, y_0) that satisfies the assumptions.
4. Generalize for functions $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Exercise 3.3 (Multi-step SDE Scheme). With the notations from the course, consider the SDE scheme defined by $Y_{n+1} = Y_n + \frac{3}{2}a_n h - \frac{1}{2}a_{n-1}h + b_n\sqrt{h}\xi_n$ where ξ_n are **discrete** i.i.d variables with mean m and finite variance σ^2 , and each ξ_n independent of the filtration $\mathcal{A}_{t_n}$. We suppose that a, b are functions independent of time, and a, b, a', b', a'', b'' bounded.

1. Find m and σ^2 such that the scheme is weakly consistent.
2. Is the scheme strongly consistent?